

Sarlink® TPV 4755B42

Teknor Apex Company - Thermoplastic Vulcanizate

Tuesday, April 25, 2023

General Information

Product Description

The Sarlink TPV 4700 Series are very high flow injection molding engineering grades with excellent UV resistance, elasticity, and surface aesthetics designed for demanding automotive applications including window encapsulation and exterior parts. Sarlink® TPV 4755B42 is a black, medium hardness, low density thermoplastic vulcanizate suited for injection molding applications that require superior flow properties.

General

Material Status	• Commercial: Active		
Availability	• Africa & Middle East • Asia Pacific	• Europe • Latin America	• North America
Features	• Chemical Resistant • Good Processability • High Flow	• Low Compression Set • Low Density • Low Specific Gravity	• Medium Hardness • Resilient
Uses	• Automotive Applications • Automotive Exterior Parts	• Automotive Window Encapsulation • Rubber Replacement	
RoHS Compliance	• RoHS Compliant		
Automotive Specifications	• DAIMLER DBL 5422 Color: Black • DAIMLER DBL 5562.30 Color: Black • FORD WSD-M2D378-A1 Color: Black	• FORD WSS-M2D378-B1 Color: Black • GM QK 003513 Color: Black • VAG VW501 23 Color: Black	• VOLKSWAGEN VW 50180 Color: Black
Appearance	• Black		
Forms	• Pellets		
Processing Method	• Injection Molding		

ASTM & ISO Properties ¹

Physical	Nominal Value	Unit	Test Method
Density / Specific Gravity	0.910		ASTM D792
Density	0.910	g/cm ³	ISO 1183
Elastomers	Nominal Value	Unit	Test Method
Tensile Stress			ASTM D412
Across Flow : 100% Strain	261	psi	
Flow : 100% Strain	392	psi	
Tensile Stress			ISO 37
Across Flow : 100% Strain	261	psi	
Flow : 100% Strain	392	psi	
Tensile Strength			ASTM D412
Across Flow : Break	725	psi	
Flow : Break	624	psi	
Tensile Stress			ISO 37
Across Flow : Break	725	psi	
Flow : Break	624	psi	
Tensile Elongation			ASTM D412
Across Flow : Break	500	%	
Flow : Break	390	%	
Tensile Elongation			ISO 37
Across Flow : Break	500	%	
Flow : Break	390	%	

Revision Date: 11/26/2019

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Elastomers	Nominal Value	Unit	Test Method
Tear Strength - Across Flow	110	lbf/in	ASTM D624
Tear Strength - Across Flow ²	108	lbf/in	ISO 34-1
Compression Set			ASTM D395
73°F, 22 hr	19	%	
158°F, 22 hr	28	%	
257°F, 70 hr	43	%	
Compression Set			ISO 815
73°F, 22 hr	19	%	
158°F, 22 hr	28	%	
257°F, 70 hr	43	%	
Hardness	Nominal Value	Unit	Test Method
Durometer Hardness			ASTM D2240
Shore A, 5 sec, Extruded	54		
Shore A, 5 sec, Injection Molded	56		
Shore Hardness			ISO 868
Shore A, 5 sec, Extruded	54		
Shore A, 5 sec, Injection Molded	56		
Aging	Nominal Value	Unit	Test Method
Change in Tensile Strength in Air - Across Flow			ASTM D573
275°F, 1000 hr	-2.0	%	
100% Strain, 275°F, 1000 hr	7.0	%	
302°F, 168 hr	-6.0	%	
100% Strain, 302°F, 168 hr	-1.0	%	
Change in Tensile Strength in Air - Across Flow			ISO 188
275°F, 1000 hr	-2.0	%	
100% Strain 275°F, 1000 hr	7.0	%	
302°F, 168 hr	-6.0	%	
100% Strain 302°F, 168 hr	-1.0	%	
Change in Ultimate Elongation in Air - Across Flow			ASTM D573
275°F, 1000 hr	-4.0	%	
302°F, 168 hr	-2.0	%	
Change in Tensile Strain at Break in Air - Across Flow			ISO 188
275°F, 1000 hr	-4.0	%	
302°F, 168 hr	-2.0	%	
Change in Durometer Hardness in Air			ASTM D573
Shore A, 275°F, 1000 hr	2.0		
Shore A, 302°F, 168 hr	-2.0		
Change in Shore Hardness in Air			ISO 188
Shore A, 275°F, 1000 hr	2.0		
Shore A, 302°F, 168 hr	-2.0		
Change in Volume (257°F, 70 hr, in IRM 903 Oil)	90	%	ASTM D471
Change in Volume (257°F, 70 hr, in IRM 903 Oil)	90	%	ISO 1817
Additional Information	Nominal Value	Unit	Test Method
Apparent Shear Viscosity			
Capillary : 392°F	220	Pa·s	ASTM D3835
Capillary @ 206/s : 392°F	220	Pa·s	ISO 11443

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Processing Information

Injection	Nominal Value	Unit
Drying Temperature - Desiccant Dryer	180	°F
Drying Time - Desiccant Dryer	3.0 to 4.0	hr
Dew Point - Desiccant Dryer	-40	°F
Suggested Shot Size	25 to 50	%
Rear Temperature	356 to 410	°F
Middle Temperature	374 to 428	°F
Front Temperature	392 to 464	°F
Nozzle Temperature	410 to 464	°F
Processing (Melt) Temp	410 to 464	°F
Mold Temperature	50 to 150	°F
Back Pressure	10.0 to 150	psi
Screw Speed	100 to 200	rpm
Screw L/D Ratio	25.0:1.0	

Notes

¹ Typical properties: these are not to be construed as specifications.

² Method Ba, Angle (Unnicked)

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